Stablecoin StressBench: An 18-Event Benchmark for Execution-Aware Stablecoin Stress

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ABSTRACT

Stablecoin StressBench begins from an 18-event catalogue of stablecoin stress from 2020–2023 across seven mechanisms: algorithmic reflexivity, fiat-reserve bank shock, regulatory winddown, exchange credit, DeFi pool imbalance, collateral liquidation, and RWA/niche stablecoin failure. We convert this historical universe into a tiered benchmark with source-verification status, coverage scores, machine-readable event windows, and claim-specific data permissions: Tier A supports VWAP labels and oracle bounds; Tier B supports price/liquidity claims; and Tier C supports taxonomy only. On the execution-grade SVB windows, price-visible arbitrage appears twelve times more often than it survives VWAP book-walking, fees, and settlement costs (34.3% optical versus 2.88% executable). Calm-trained models fail economically despite richer features and sequence models, while cross-mechanism meta-labeling trained on Terra/LUNA recovers 51.0% of the oracle on SVB (+82.5 bps), and a four-event mechanism-diverse protocol reaches 51.8% (+83.7 bps). A conditioned PPO-GRU agent at identical training density earns −29.2 bps, showing that reward-only policy learning does not recover the executable subset without explicit binary supervision. We release the event catalogue, source-verification registry, execution labels, oracle, model scripts, and paper tables.

KEYWORDS

stablecoin arbitrage, market microstructure, execution-aware labels, financial benchmarks, cryptocurrency

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1 INTRODUCTION

Stablecoin stress is not a single-mechanism phenomenon. From 2020 to 2023, StressBench identifies 18 episodes spanning algorithmic death spirals, reserve-bank shocks, regulatory winddowns, exchange-credit collapses, DeFi pool imbalances, collateral-liquidation failures, and niche collateral failures. These events differ in trigger, duration, venue, affected asset, and data observability. The benchmark problem is therefore not merely to detect a depeg, but to determine which historical stress signatures remain economically executable across mechanisms.

Our thesis is that stablecoin stress is historically mechanism-diverse, but executable arbitrage is microstructure-scarce. When Silicon Valley Bank collapsed in March 2023, USDC briefly depegged and cross-venue price gaps appeared profitable on every chart. Almost none of them could be traded at a profit. VWAP slippage consumed the spread as orders cleared depth, taker fees eroded the margin, and centralised-exchange (CEX) settlement latency turned a profitable window into a loss before the transfer confirmed. Only 2.88% of one-minute windows survived all three costs and remained net-profitable. We call dislocations that appear on price charts but vanish under execution optical, and the surviving fraction executable. Every prior stablecoin benchmark uses the optical layer to define labels—which means every published performance number overstates what a model can actually achieve, because it is measured against windows that cannot be traded at a profit. The only empirically validated path to capturing executable dislocations is training on a mechanistically distinct stress event. We show that a model trained on an algorithmic crypto collapse (Terra/LUNA) transfers to a traditional bank run (USDC/SVB) because CEX market makers withdraw depth under uncertainty regardless of the trigger's cause—the order book does not know why liquidity is leaving, only that it is.

Stablecoin StressBench corrects this by building a historical-to-execution benchmark. The paper follows one spine: an 18-event historical stress universe, a mechanism taxonomy, a data-tier funnel, an execution-aware benchmark, cross-mechanism transfer, and an RL failure diagnostic. The catalogue defines the event universe; the tier system separates price/liquidity-grade evidence from execution-grade windows; VWAP book-walking creates per-minute executable labels; and the model ladder asks which historical stress signatures transfer to an unseen reserve-bank shock. The remainder of the paper answers four research questions: what the historical stress universe looks like, how much apparent arbitrage survives execution, whether mechanism diversity helps transfer, and why reward-only RL fails.

Contributions.

(1) Historical catalogue. We provide a source-tracked 18-event stablecoin stress catalogue across seven mechanisms, with event-level claim provenance, verification flags, coverage scores, machine-readable windows, and source-verified paper claims.
(2) Data-tier benchmark. We formalise Tier A/B/C claim permissions so the benchmark separates execution-grade windows, price/liquidity-grade events, and context-grade taxonomy.

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Table 1: Released StressBench artefacts.

Component	Contents	Role
Historical catalogue	18 events, mechanisms, dates, tiers	defines stress universe
Source registry	claim, source type, verification flag, paper-use flag	auditable claims
Event windows	machine-readable YAML windows	reproducible splits
Execution labels	VWAP net-profit labels, depth provenance	tradable target
Oracle	hindsight executable ceiling	upper bound
Model harness	rules, ML, meta-labeling, PPO-GRU	reproducible baselines
Paper tables	frozen CSV outputs	result verification

(3) Execution labels. We construct VWAP book-walking labels with route provenance, taker fees, settlement penalties, and a hindsight oracle ceiling.

(4) Empirical result. We show a 12× optical-to-executable gap and a notional-scaling collapse: at $10K, 34.3% of SVB minutes are optically dislocated but only 2.88% are executable.

(5) Transfer and policy result. Cross-mechanism meta-labeling recovers 51.0–51.8% of the oracle, while a conditioned PPO-GRU earns −29.2 bps despite identical positive-label density.

2 RELATED WORK

2.1 Stablecoin Stress Events

Research on stablecoin stress events defines the empirical setting for the benchmark and supplies the out-of-distribution event we transfer from. Uhlig [1] analyses the Terra/LUNA algorithmic collapse, in which an algorithmic redemption mechanism created a death-spiral run on the UST stablecoin, ultimately destroying approximately $40B in market capitalisation over three days. We use Terra/LUNA as our cross-mechanism training split precisely because its trigger is mechanistically different from the SVB bank shock: algorithmic reflexivity versus fiat-reserve withdrawal are distinct failure modes, so a model that transfers between them has learned execution structure rather than event-specific noise. Griffin and Shams [2] document that quoted stablecoin prices can reflect non-fundamental demand, which motivates our distinction between optical dislocations (visible on price charts) and executable ones (profitable after order-book costs). Hernandez Cruz et al. [21] study DEX liquidity during the SVB collapse but stop short of modelling execution costs on centralised exchanges. None of this prior work provides per-minute executable labels or a hindsight oracle ceiling against which model performance can be calibrated.

2.2 Limits to Arbitrage and Crypto Fragmentation

The limits-to-arbitrage literature supplies the economic logic that our per-minute labels turn into direct measurements. Shleifer and Vishny [4] and Gromb and Vayanos [5] establish the theoretical foundation: even when a price gap is visible, capital constraints and execution frictions prevent arbitrageurs from closing it fully, so the gap can persist. Makarov and Schoar [7] bring this argument to cryptocurrency, showing empirically that cross-venue price gaps disappear once order-book depth and settlement latency are accounted for. Our 12× ratio is the same effect measured at the one-minute label level during a stress event. Hautsch et al. [3] quantify settlement latency as a first-order execution cost on blockchain assets, which we implement as the fixed δ=5 bps penalty in our net-profit label (validated empirically in §7 using on-chain gas data). Cont and Kukanov [6] establish that the correct object for measuring execution profitability is the VWAP price achieved by walking the limit-order book, not the quoted mid-price. Our book-walking model implements this directly. Gogol et al. [18] measure a Maximal Arbitrage Value for AMM-to-CEX routes. Our oracle extends this concept to stress-event dislocations on centralised exchanges and to model-evaluation benchmarking. Where prior theoretical work characterises the existence of execution frictions, we measure them at one-minute resolution with empirical order-book depth.

2.3 Financial Machine Learning and RL Execution

Our modelling choices build directly on the financial machine-learning and reinforcement-learning execution literatures, which we both adopt and extend. López de Prado [8] introduces meta-labeling, a two-stage framework in which a primary signal identifies candidate opportunities and a secondary classifier learns when those opportunities are genuine. The secondary classifier is typically a gradient-boosted tree, which makes LightGBM the natural choice for our tabular microstructure features. We extend meta-labeling to the cross-mechanism setting, training on Terra/LUNA and evaluating on SVB. Four training conditions establish that oracle capture is stable near 50% and mechanism-independent. Cintra and Holloway [9] demonstrate cross-event transfer of Bayesian Online Changepoint Detection [15] using AMM reserve depletion signals, detecting the Terra depeg twelve hours early. We replicate their setup on centralised order-book data and find the result does not transfer: BOCPD achieves AUROC 0.229 on CEX cross-quote data because CEX basis is mean-reverting rather than gradually depleting. This negative replication is itself a contribution: it rules out changepoint detection as a substitute for the training-distribution fix that meta-labeling provides, and confirms that AMM-to-CEX transfer of detection methods does not hold. On reinforcement learning, Moallemi and Wang [16] and Ning et al. [17] establish the finite-horizon MDP

Table 2: Route-leg depth provenance.

Leg	SVB Mar 2023	2024 windows
Buy BTC-USDC	kline-proxy†	real L2 (futures)
Cross USDC-USDT	partial/proxy; real L2 where archived	real L2 (futures)
Sell BTC-USDT	raw/futures depth	real L2 (futures)
Ref BTC-USD	candle-proxy	candle-proxy

†BTC-USDC perpetual listed Jan 2024; all 2022 windows kline-proxy.

framing for optimal order execution that we adopt in §5.2. Mohl et al. [22] demonstrate that GPU-scale multi-agent RL is feasible on production limit-order-book data, training on approximately 400 million real orders for a joint execution and market-making task. Olby et al. [23] construct a reactive simulator that updates book depth in response to agent order flow, benchmarking RL timing policies against an Almgren–Chriss efficient frontier. Both of these works operate in environments where the agent's actions affect the order book in real time. Our diagnostic operates in the complementary setting of historical replay without market-impact feedback, and the contrast is what lets us isolate training supervision, rather than environment reactivity, as the binding constraint.

3 BENCHMARK DESIGN

3.1 Data Sources and Instruments

The final dataset contains 56,134 1-minute bars across six windows (Table 3). Price features are sourced from Binance Vision (klines and aggTrades) and Coinbase REST candles. Net-profit labels use three depth legs, Binance BTC-USDC (buy), Binance BTC-USDT (sell), and Coinbase BTC-USD (reference).

Data quality varies by leg and window, and every depth record is tagged with a provenance field (Table 2). The sell leg (BTC-USDT) and cross leg (USDC-USDT) use real L2 or raw futures bookDepth where archived, with proxy/partial tags where route coverage is incomplete. The buy leg (BTC-USDC) is the main challenge. The BTC-USDC perpetual contract did not exist on Binance until January 2024, so the SVB window requires kline-proxy depth, one-minute candlestick volume as a substitute for unarchived order-book depth (high-volume candles do not guarantee deep books, and a 20% haircut bounds this error). Coverage is tracked at the row and route-leg level rather than silently imputed. For Mar 10–11, incomplete BTC-USDC and USDC-USDT route legs are tagged proxy/partial; Mar 12–20 has complete cross-leg coverage in the committed route audit. All headline execution claims are recomputed under the 20% depth haircut, and Figure 1 reports the route-leg coverage audit used to separate execution-grade SVB claims from validation/training-grade Terra/LUNA labels. For all 2024 windows, real L2 data is available. To bound the proxy error, we apply a 20% depth haircut to the kline-proxy buy leg. The executable rate falls from 2.88% to 2.40%, the ratio widens from 12× to 14×, and the oracle remains above +130 bps. The headline result is therefore robust to the approximation.

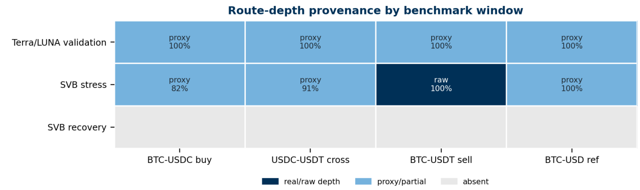

Figure 1: Route-depth provenance by benchmark window. Dark cells are real/raw depth, blue cells are proxy or partial route coverage, and grey cells are absent.

Table 3: Dataset splits (56,134 1-min bars).

Split	Event	Min.	Pos.
Train	Calm control (×3)	28,776	0.00%
Validation	Terra/LUNA May 2022	11,526	2.30%
Test	USDC/SVB Mar 2023	15,832	2.88%

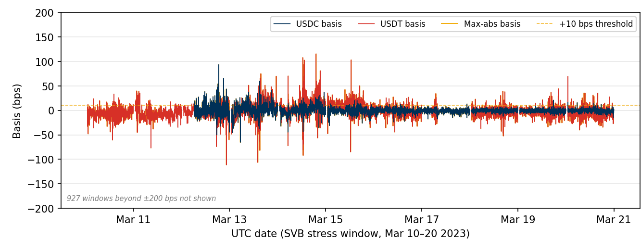

Figure 2: Cross-quote basis (bps) during the SVB test window (Mar 10–20 2023). USDC basis (navy): 12.4% of minutes > 10 bps; USDT basis (red): 27.4%; max-abs (gold): 34.3%. The price-to-execution gap is 12×.

3.2 Event-Based Split Design

Splits are defined by event identity rather than time, ensuring that no model trained on the train split has encountered stress dynamics (Figure 2). Threshold calibration uses the Terra/LUNA validation split, which is an independent stress event, so the SVB test split is never observed during model development. Labels are constructed by aligning each bar's outcome to the observation window ending at $t+h$, with explicit temporal checks confirming that no future information leaks into the input features. The 2.88% positive rate in the test split implies severe class imbalance. We therefore report economic net bps as the primary criterion rather than AUROC or AUPRC, which are secondary metrics.

3.3 Historical StressBench: Event Universe and Mechanism Coverage

Stablecoin StressBench begins with the historical event universe, not the SVB case study. The catalogue records 18 stress events from 2020–2023 and assigns each event a mechanism class, source-verification status, coverage score, event window, tier, and permitted empirical use. It is used to define the event universe, select cross-mechanism training events, assign claim types, and separate price-grade from execution-grade tasks.

Table 4: Historical StressBench event matrix. The catalogue covers 18 events across seven mechanisms; tier ranges define permitted claims, not equal execution coverage.

Mechanism class	N	Example	Tier range	Role in paper
Algorithmic/reflexive	5	Terra/UST	B–C	validation, transfer source
Fiat-reserve bank shock	2	USDC/SVB	A	execution benchmark
Regulatory winddown	2	BUSD	B–C	transfer stress mechanism
Exchange credit/liquidity	3	FTX, Celsius/3AC	B	transfer and contrast
DeFi pool imbalance	3	Curve/UST, USDT/Curve	B	on-chain liquidity stress
Collateral/liquidation	1	DAI Black Thursday	B	above-peg stress
RWA/niche	2	Acala aUSD, USDR	B–C	long-tail stress coverage
Total	18		2A, 11B, 5C	historical stress universe

3.3.1 *Event inclusion criteria.*

An event enters the catalogue if it produced a material stablecoin peg, liquidity, redemption, or collateral stress episode and has enough documentation to assign a trigger mechanism and event window. Exact magnitudes enter the paper only when the source-verification registry marks the claim as verified and paper-usable; otherwise the event contributes mechanism coverage and estimated magnitudes are marked as such.

3.3.2 *Mechanism classes.*

Table 4 summarises the seven mechanism classes. The table is deliberately placed before the model section because it defines the benchmark population from which transfer experiments are drawn.

3.3.3 *Data-tier funnel.*

The catalogue is filtered by data quality before it becomes an execution benchmark (Figure 4). The resulting three-layer distinction is central to the paper: the historical catalogue defines the 18-event stress universe; the historical empirical panel contains events with usable price or liquidity rows; and the execution benchmark contains Tier-A VWAP-labelled windows. The execution benchmark is a strict subset of the empirical panel, which is a strict subset of the catalogue.

3.3.4 *Source verification and claim registry.*

Every event-level claim is tied to a source type, URL/status, verification flag, and paper-use flag in the benchmark registry. Table 5 states the claim boundary used in the paper. This prevents Tier-B and Tier-C events from being cited as if they had route-complete order-book evidence.

3.3.5 *How the catalogue determines model splits.*

The 18-event catalogue determines the transfer design. Terra/LUNA supplies algorithmic/reflexive stress, Celsius/3AC and FTX supply exchange-credit/liquidity stress, and BUSD supplies regulatory winddown stress. SVB is held out as the fiat-reserve bank-shock test event. The pooled transfer experiment is therefore not a generic data augmentation exercise; it tests whether execution microstructure learned from distinct stress mechanisms transfers to an unseen reserve-bank shock.

Table 5: Event claims and permitted empirical use.

Tier	Events	Computable outputs	Used for
A	2	VWAP net profit, oracle, model P&L	execution benchmark
B	11	price basis, depeg magnitude, liquidity proxies	mechanism transfer / stress contrast
C	5	mechanism classification	historical coverage

Table 6: Comparison with related benchmarks and datasets. ✓= fully provided; ~ = partially; — = absent.

Work	Stress event	Exec. labels	L2 prov.	Oracle	Cross-event
Cintra & Holloway [9]	✓	—	AMM	—	✓
Hernandez Cruz et al. [21]	✓	—	DEX	—	—
Gogol et al. [18]	—	~	CEX	MAV	—
Adams et al. [19]	—	~	CEX	—	—
StressBench (ours)	✓	✓	CEX	✓	✓

3.4 Benchmark Positioning

Table 6 positions StressBench against the closest prior work along five axes. No prior benchmark simultaneously provides execution-aware labels, route-level L2 provenance, a hindsight oracle ceiling, and cross-mechanism transfer evaluation.

Unlike execution simulators that optimise within known LOB environments, StressBench studies cross-mechanism transfer under sparse historical stress labels and fixed replayed liquidity.

3.5 Feature Sets

Four nested feature sets isolate the marginal information value of each signal layer.

Feature set	Description
`price_only`	5 cross-quote basis columns
`price_plus_book`	+7 microstructure (spread, depth, imbalance)
`price_book_frag`	+3 cross-venue fragmentation columns
`price_book_settle`	+7 on-chain settlement proxies

4 EXECUTION-AWARE LABELS

Labels operationalise the three-layer framework from §1. The price-to-execution gap is the ratio of Layer 1 (optical) minutes to Layer 2 (executable) minutes and quantifies how misleading price-based benchmarks are. The oracle gap is the performance difference between Layer 2 in hindsight and Layer 3 (predictable with a real model), quantifying how hard the prediction problem is even with the correct labels. Both gaps are nonzero, independent, and large. This section details how they are constructed.

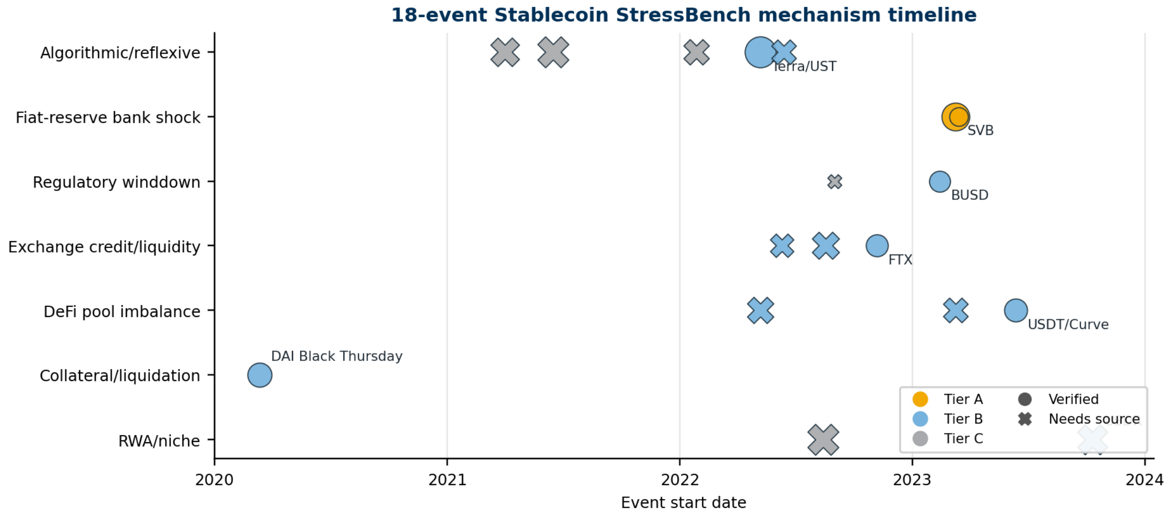

Figure 3: The 18-event mechanism timeline. Marker size scales with absolute depeg magnitude; colour gives Tier A/B/C; marker shape distinguishes verified claims from events that need additional source confirmation.

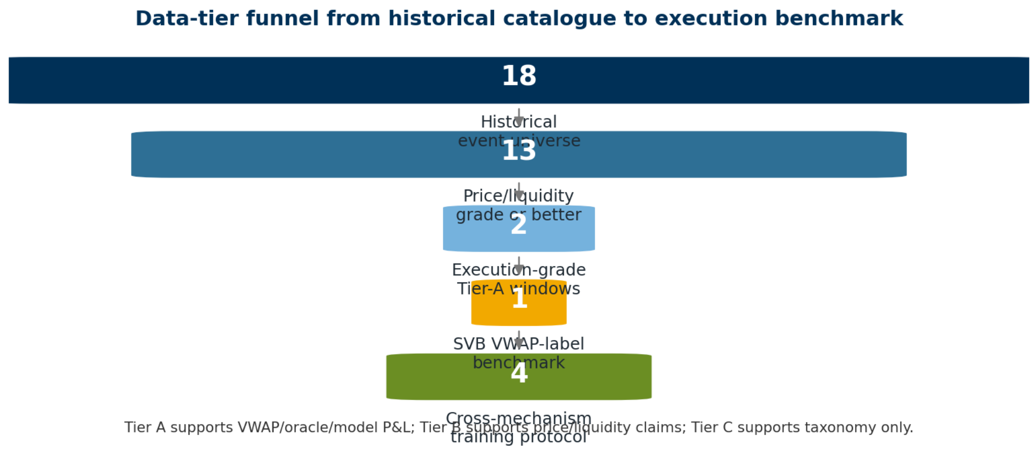

Figure 4: Data-tier funnel from the 18-event catalogue to the execution benchmark and transfer protocol. Tier A supports VWAP labels and oracle bounds; Tier B supports price/liquidity claims; Tier C supports historical taxonomy only.

4.1 Cross-Quote Basis and Labels

The basis measures the percentage difference between BTC priced in USDC and BTC priced in real dollars. A non-zero basis signals an apparent dislocation. Three equations define the label construction.

$$b_{\mathrm{USDC}}(t) = 10{,}000 \times \frac{P_{\mathrm{BTC/USDC}}(t) - P_{\mathrm{BTC/USD}}(t)}{P_{\mathrm{BTC/USD}}(t)} \quad [\mathrm{bps}] \tag{1}$$

$$b_{\max}(t) = \max(|b_{\mathrm{USDC}}(t)|,\ |b_{\mathrm{USDT}}(t)|) \tag{2}$$

$$y_{\mathrm{basis}}(t) = \mathbf{1}[|b_{\mathrm{USDC}}(t{+}h)| > \tau] \tag{3}$$

Equations (1)–(3) define the basis, the pooled signal, and the label. Primary task: `basis_usdc_1m_gt10bps` (τ=10 bps, h=1 min).

4.2 VWAP Execution and Net Profit Labels

$$\mathrm{net}(q,t) = 10{,}000 \times \left[\frac{\mathrm{VWAP}_{\mathrm{sell}} - \mathrm{VWAP}_{\mathrm{buy}}}{P_{\mathrm{ref}}} - f_{\mathrm{buy}} - f_{\mathrm{sell}} - \delta_{\mathrm{settle}} \right] \quad [\mathrm{bps}] \tag{4}$$

$$y_{\mathrm{arb}}(q,h,t) = \mathbf{1}\left[\max_{s \in [t,\, t+h]} \mathrm{net}(q,s) > c \right] \tag{5}$$

Here P_{ref} is the Coinbase BTC-USD mid-price and $f_{\mathrm{buy}}, f_{\mathrm{sell}}, \delta_{\mathrm{settle}}$ are per-leg taker fees and settlement latency, all as dimensionless fractions. The profit threshold is c=10 bps, taker fee 4 bps per side, and δ=5 bps. Equations (4) and (5) define the primary economic task `executable_arb_q10000_5m`. The hindsight oracle takes every minute where $\mathrm{net}(q,t) > c$, establishing a ceiling no deployable model can exceed.

Table 7: Model ladder and expected failure modes.

Model family	Purpose	Expected failure mode
Price thresholds	prior price-only arbitrage proxy	false positives
Tabular ML	execution-label classifier	no stress positives in calm training
Sequence models	temporal depth withdrawal	ranking without P&L
Meta-labeling	cross-mechanism executable filter	partial oracle capture
PPO-GRU	reward-only policy learning	overtrading

Table 8: Feature ablation under calm-trained LightGBM (`basis_usdc_1m_gt10bps`).

Feature set	Net bps	Trades	AUROC	AUPRC	Oracle cap.
`price_only`	−125.3	487	0.663	0.248	−77.3%
`price_plus_book`	−101.5	217	0.580	0.216	−62.6%
`price_book_frag`	−101.5	217	0.580	0.216	−62.6%
`price_book_settle`	−101.5	217	0.580	0.216	−62.6%

5 MODELS AND EVALUATION

5.1 Model Ladder

The ladder is organised around the benchmark thesis: price-visible stress is common, but executable opportunities are scarce and require mechanism-aware supervision. The two anchors, `NoTrade` (zero net bps) and the hindsight `Oracle`, bracket the achievable range; every other result is interpreted as a fraction of the distance between them. Table 7 compresses the model families and their intended failure modes.

The price-threshold rules reproduce the implicit labels of prior optical-basis work [11]. Tabular classifiers and four nested feature sets test whether adding microstructure, fragmentation, and settlement signals closes the gap. GRU, MLP-Window, EWMA, CUSUM, and BOCPD test whether temporal or changepoint structure substitutes for stress training. Meta-labeling [8] and PPO-GRU [16, 17] then isolate supervision format: same Terra/LUNA density, explicit binary labels versus sparse reward.

Microstructure features matter only after stress exposure is present. Under calm training, richer features do not rescue P&L; under cross-mechanism training, depth and spread features become predictive because the model has seen stress-regime depth withdrawal.

5.2 RL Execution Policy Baseline

This subsection asks whether sequential decision-making can learn a profitable policy given the correct training density, by framing the entry-timing problem as a finite-horizon Markov Decision Process (MDP). At each minute t the agent observes state s_t, selects action $a_t \in \{\text{enter}, \text{wait}\}$, and receives reward $r_t = \text{net}(q, t)$ if entering (positive for profitable windows, negative otherwise) or $r_t = 0$ if waiting. The state s_t is the 30-minute look-back window of `price_plus_book` features, the smallest feature set that delivers meaningful microstructure lift over price-only signals (see §6.4 for SHAP evidence). This feature set is available in real time without settlement-latency dependency. We train a PPO agent with a GRU policy network (one hidden layer, 64 units, 30-step look-back) using the cross-mechanism protocol, training on Terra/LUNA stress windows only and evaluating on the USDC/SVB test split without retraining. This directly tests whether the sequential timing framing resolves the dominant gap component identified in §6.3.

5.3 Calibration and Evaluation

Net bps denotes realised mean net basis points per triggered trade after VWAP book-walking, taker fees, and settlement penalties; zero trades are reported as NoTrade rather than an undefined average. Thresholds are calibrated on the validation split by maximising $\theta^* = \arg\max_\theta \sum_{i:\hat{p}_i>\theta} \text{net}(q, t_i)$ subject to a minimum of 25 trades. For cross-mechanism meta-labeling, Terra/LUNA is treated as a blocked training-and-calibration event: the secondary classifier is fit on the training block, the decision threshold is selected on the calibration block, and SVB remains untouched until final evaluation. The primary metrics are mean net bps per triggered trade and oracle capture percentage, defined as model mean net bps divided by oracle mean net bps (the fraction of the oracle's per-trade return that a model recovers). Secondary metrics are AUROC (area under the ROC curve) and AUPRC (area under the precision-recall curve). `NoTrade` at 0 bps is the economic floor.

All results are reported on the SVB test split (N=15,832) with the cost parameters stated above. The kline-proxy approximation on the BTC-USDC buy leg is bounded by the haircut sensitivity in §6.2. Five explicit checks guard against leakage and evaluation artefacts:

(1) Random label shuffles produce zero or negative P&L for all models.
(2) Shifting features forward by one minute to simulate lookahead does not improve performance.
(3) The calm-control training split contains exactly zero executable arbitrage windows.
(4) Train and test events are separated by event identity; no SVB features appear during model development.
(5) Cross-mechanism threshold calibration uses only the held-out Terra/LUNA calibration block; the SVB test split is held out until final evaluation.

Table 9 summarises the five evaluation tasks shipped with StressBench v1.0.

A valid StressBench submission must report mean net bps per triggered trade, trades, hit rate, oracle capture, AUROC/AUPRC, cost assumptions, notional size, calibration split, and whether the model uses Tier-A, Tier-B, or source-tracked events. Models are ranked by net bps and oracle capture, not by AUROC alone.

Table 9: StressBench v1.0 benchmark card.

Task	Train	Val.	Test	Primary metric
optical_basis	calm	Terra	SVB	AUROC, AUPRC
exec_arb_q10k	calm	Terra	SVB	net bps, oracle cap.
exec_arb_q50k	calm	Terra	SVB	net bps
cross_mech	Terra block	Terra calib.	SVB	net bps, oracle cap.
policy_entry (RL)	Terra block	Terra calib.	SVB	net bps, trades

6 RESULTS

6.1 RQ1: Historical Stress Universe and Data-Tier Funnel

The historical benchmark covers 18 events, seven mechanisms, two execution-grade Tier-A windows, eleven price/liquidity-grade Tier-B events, and five context-grade Tier-C events (Table 4, Figure 3). The data-tier funnel is the key design choice: all events define the stress universe, only price/liquidity-grade events support historical stress summaries, and only Tier-A windows support VWAP labels, oracle bounds, and model P&L (Figure 4). This makes the narrowing process benchmark-driven rather than case-study-driven.

6.2 RQ2: The 12× Execution Barrier

The first gap is between what a price chart shows and what an order book will fill. At a 10 bps threshold, 34.3% of test minutes are optically dislocated but only 2.88% are executable after VWAP slippage, taker fees, and settlement latency. Twelve times more arbitrage appears than can be traded, and the ratio widens further with institutional trade size (Table 12). This ratio is not a threshold artefact and persists across all tested basis levels and is robust to data-quality concerns. Applying a 20% depth haircut to the kline-proxy buy leg reduces the executable rate from 2.88% to 2.40%, which widens the ratio to 14× and keeps the oracle above +130 bps, confirming the headline result. The gap also deepens with institutional scale. Figure 6 shows the executable rate falling from 2.88% at $10K to 0.03% at $500K, with the steepest drop between $10K and $50K. Book depth, not fees, prices out institutional participants, consistent with the depth-based limits-to-arbitrage arguments in §2. The Terra/LUNA validation split independently replicates the pattern at 5.9×, providing directional evidence that the gap is not SVB-specific. Table 10 shows the stress/recovery split: all executable windows are in the stress phase, and the recovery phase contains zero despite 21.6% optical activity.

Optical dislocations are a consistent feature of stablecoin stress, appearing in all seven mechanism classes with optical positive rates ranging from 4.0% (BUSD, regulatory winddown) to 53.0% (USDC/SVB stress phase, Table 10). The 12× execution gap is empirically verified only for Tier-A events where route-complete L2 depth is available, but the universal presence of optical dislocations across mechanisms establishes that the phenomenon StressBench measures is not SVB-specific.

The committed result tables store model-level summaries but not every model's per-minute trade decisions, so model-return confidence intervals are reserved for the reproducibility appendix once those decision traces are frozen.

The gap is robust across the full cost-parameter space: sweeping taker fees from 2 to 10 bps per side and settlement penalty from 0 to 10 bps moves the executable rate only between 2.84% and 3.66%, and the ratio to the optical rate

Table 10: Optical and executable diagnostics for SVB execution-grade windows and the Terra/LUNA validation window. SVB stress/recovery are Tier-A execution-grade; Terra/LUNA is validation/training-grade and is not used for final execution-grade claims.

Window	Min.	Opt.	Exec.	Oracle
SVB Stress (Mar 10–14)	7,189	53.0%	6.63%	+259.9 bps
SVB Recovery (Mar 15–20)	8,643	21.6%	0.00%	—
Terra/LUNA (May 2022, val.)	11,526	14.0%	2.40%	+87.6 bps

Opt. = max-abs basis > 10 bps; Exec. = net-profitable windows.
Terra/LUNA labels are used for cross-mechanism training and calibration only.

Table 11: Mechanism-level stress-intensity hypotheses. Only the fiat-reserve bank-shock row has execution-grade evidence; all other rows define observable stress channels and permitted paper use.

Mechanism	Tier	Observable stress	Execution evidence	Paper use
Fiat-reserve bank shock	A	price + L2 + labels	verified	execution benchmark
Algorithmic/reflexive	B	price/liquidity	training/validation only	transfer source
Exchange credit	B	price/liquidity	no final execution claim	transfer/contrast
Regulatory winddown	B	price/liquidity	no final execution claim	transfer
DeFi pool imbalance	B	pool/liquidity	no final execution claim	mechanism context
Collateral liquidation	B	price/liquidation	no final execution claim	above-peg stress
RWA/niche	B/C	price/context	no final execution claim	long-tail coverage

Table 12: Price-to-execution gap ($10K, 1-min horizon, test split).

Thresh.	P_{max}	P_{USDC}	Exec.	Ratio
0 bps	95.86%	82.96%	3.34%	29×
5 bps	61.99%	25.32%	3.03%	20×
10 bps	34.33%	12.45%	2.88%	12×
25 bps	9.54%	6.48%	2.62%	3.6×

Table 13: Block-bootstrap uncertainty for directly auditable SVB claims (60-minute blocks, 2,000 resamples).

Claim	Point	95% CI
Optical rate	34.3%	[31.2, 37.5]%
Executable rate	2.88%	[1.47, 4.59]%
Optical/executable ratio	11.9×	[7.6, 23.0]×
Oracle mean net bps	+259.9	[+213.9, +304.2]

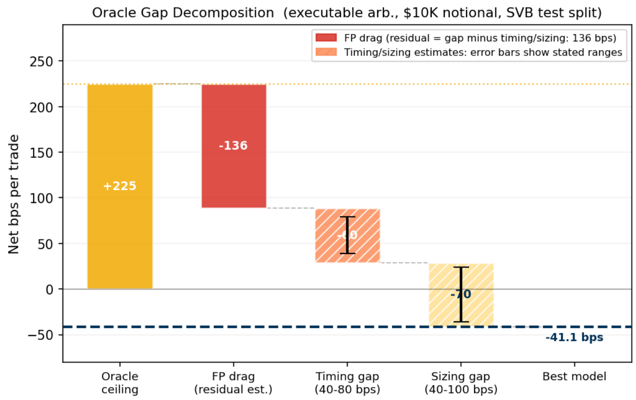

Figure 5: Oracle gap decomposition ($10K notional). Three components account for the 266 bps gap from oracle (+224.6 bps) to best model (−41 bps): FP drag, timing error, and sizing error.

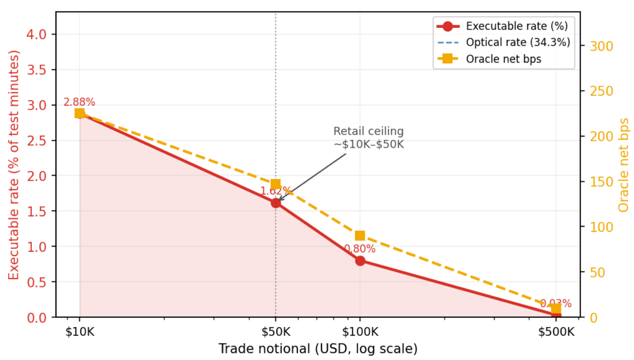

Figure 6: Notional scaling (SVB test split). The executable rate collapses from 2.88% at $10K to 0.03% at $500K: book depth, not fees, is the binding institutional constraint.

remains above 3.5× throughout. Claim boundaries and data-tier scope are stated in §5.3.

6.3 RQ3: Prediction Gap Under Calm Training

The second gap is between a profitable window and a model that can find it in advance. Even with execution-aware labels, every model trained on calm data loses money on the executable task. The hindsight oracle earns +224.6 bps while the best deployable model reaches −41.1 bps, a structural gap of more than 260 bps that richer features and sequence models do not close (Table 14; Figure 7).

Price-threshold rules overtrade massively, generating 607 to 1,969 trades at −136 to −347 bps. Their mean return when wrong (−294 bps to −370 bps) reflects near-total false-positive exposure. Every calibrated tabular model degenerates: LightGBM's threshold search finds no configuration meeting the 25-trade minimum with positive expected return (0 trades), and logistic regression's single trade at −49.1 bps is not statistically meaningful. The 2.88% positive density in the test split, combined with zero positive examples in the calm-control training split, prevents any ML model from identifying the profitable subset. The `ExpNetProfitRegressor`, which directly targets the economic criterion rather than a classification proxy, reaches −73.8 bps, ruling out label mismatch as the explanation. The gap is structural (Table 15). GRU and MLP-Window sequence models confirm this, with AUROC 0.80 yet −239 bps (GRU) and −341 bps (MLP-Window). High ranking accuracy does not translate to positive returns when 97% of test windows are non-executable and the training split contains zero stress examples.

Table 14: Oracle gap and best-model net bps (USDC/SVB test split). *Best calibrated model is NoTrade (zero trades). ‡30-min GRU window.

Task / Model	Oracle	Best model	Gap	AUPRC
`basis_usdc_1m`	+161.7 bps	0 bps*	162 bps	0.253
`exec_arb_q10k_5m`	+224.6 bps	−41.1 bps	266 bps	0.060
`exec_arb_q50k_5m`	+146.8 bps	−112.7 bps	260 bps	0.063
GRU (seq)‡, `basis_usdc_1m`	+161.7 bps	−239.0 bps	401 bps	—

Kline-proxy buy leg; 20% haircut sensitivity in §6.2.

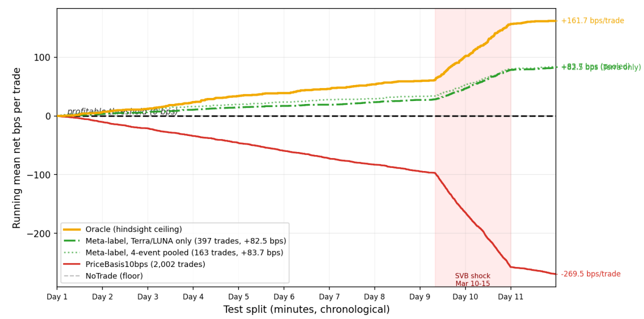

Figure 7: No calm-trained model ever crosses zero. Meta-labeling trained on a structurally different event (Terra/LUNA) is the only approach that achieves positive returns, earning +82.5 bps—roughly half the oracle's +161.7 bps. Every other learned model loses money despite sometimes high AUROC. Oracle (gold); meta-label Terra/LUNA only (green solid); meta-label 4-event pooled (green dotted); PriceBasis10bps (red); NoTrade (grey dashed). Shaded: SVB shock window.

Table 15: Tabular model ladder (**`basis_usdc_1m_gt10bps`**). Hit%: fraction of trades net-profitable. †Oracle ceiling. *0-trade degenerate: no threshold meets the 25-trade minimum with positive expected return.

Model	Feat.	Net bps	Trades	AUROC	Hit%
Oracle†	—	+161.7	316	—	100%
`lgbm`*	all	—	0	0.653	—
`logistic`	all	−49.1	1	0.133	0%
`gross_arb`	px	−136.0	611	0.531	4%
`price_10`	px	−269.5	2002	0.833	6%
`price_25`	px	−347.3	1025	0.746	5%
`no_trade`	—	0.0	0	—	—
`ExpNetProfit`	px	−73.8	537	—	3%

px = `price_only`; all = `price_book_settle`.

All calibrated ML models degenerate to zero trades under calm-training, equivalent to NoTrade. This section examines why the prediction gap is structural; the cross-mechanism fix that resolves it is in §6.4. Figure 5 decomposes the 260 bps gap into three measurable components. False-positive exposure is the dominant drag. `PriceBasis10bps` generates 1,969 trades, of which 1,851 are false positives. A root-cause analysis identifies route-direction mismatch as the primary failure mode: 581 of those false positives are triggered by USDT dislocations where the USDC route is simultaneously non-executable, at −38.7 bps each. True-positive windows carry a mean USDC basis of 344 bps, while false-positive windows carry just 0.56 bps and exhibit higher average bid depth (72,805 vs. 59,961 BTC units), ruling out thin books as the main failure. The remaining gap splits into timing error (40–80 bps from entering 1–2 minutes late in a declining depth regime) and sizing error (40–100 bps as depth depletion shrinks the executable fraction from 2.88% at $10K to 1.62% at $50K). Meta-labeling's approximately 51% hit rate versus 6% for the price rule shows that cross-mechanism training substantially reduces route-direction mismatch, though a 49% false-positive rate and roughly 113 missed profitable windows remain, confining net returns to +82.5 bps.

6.4 RQ4: Cross-Mechanism Transfer and Policy Failure

The third gap is between a correct signal and a profitable policy. Cross-mechanism meta-labeling trained on Terra/LUNA is the only method that crosses zero, earning +82.5 bps and recovering half of the oracle return. A reinforcement learning agent given the same 17% training density earns −29.2 bps, which isolates explicit binary supervision as the ingredient that reward optimisation cannot replicate at low positive rates.

6.4.1 Cross-Mechanism Transfer and Training Diversity. Cross-mechanism meta-labeling trained on Terra/LUNA achieves +82.5 bps on SVB (51.0% oracle capture), while calm-control training yields zero meta-positives by construction. Retraining on Terra/LUNA (1,559 minutes where the primary threshold triggers, 265 meta-positives, 17% positive rate) and evaluating on SVB, with all fitting and threshold selection confined to Terra/LUNA data, produces 397 trades versus zero with calm training (Table 17). Table 16 states how the 18-event catalogue selects the transfer events.

Figure 9 provides direct empirical proof: ask-side depth collapses and spread widens identically in primary-signal windows across Terra/LUNA (algorithmic) and SVB stress (fiat-reserve), explaining why training on one event transfers to the other. The reason these features are predictive is visible in the SHAP decomposition: price features carry 94.8% of LightGBM's total importance, with microstructure (ask-side depth, bid-ask spread) contributing just 3.4% combined. A model trained on calm data learns calm price dynamics. The microstructure features are present but uninformative because the training distribution contains no stress

Table 16: Transfer-event selection from the 18-event catalogue.

Event	Mechanism	Role	Reason
Terra/LUNA	algorithmic/reflexive	validation/train	high stress signal
Celsius/3AC	exchange credit	train	liquidity contagion
FTX	exchange credit	train/contrast	low-intensity shock
BUSD	regulatory winddown	train	policy winddown
SVB	fiat-reserve shock	held-out test	execution-grade anchor

Table 17: Cross-mechanism meta-labeling results (**`basis_usdc_1m_gt10bps`**, **`price_plus_book`**).

Training split	Net bps	Trades	Oracle cap.
Oracle (ceiling)	+161.7	316	100.0%
Terra/LUNA (cross-mech.)	+82.5	397	51.0%
Calm-control (baseline)	0.0	0	0.0%

Primary signal $|b_{\mathrm{USDC}}|$>10 bps; threshold calibrated on validation.

Table 18: Mechanism-diverse transfer results (**`basis_usdc_1m_gt10bps`**, **`price_plus_book`**).

Training source	Mechanism	Net bps	Trades	Oracle cap.	Interpretation
Calm control	none	0.0	0	0.0%	no stress positives
Terra/LUNA	algorithmic	+78.5	123	48.4%	depth withdrawal transfers
Celsius/3AC	exchange credit	+79.7	221	49.1%	liquidity-contagion signal
FTX	exchange credit	+36.5	1	22.5%	low intensity
BUSD	regulatory	—	—	—	pooled-only stress source
Pooled four-event	multi-mechanism	+83.7	163	51.6%	diversity stabilises transfer

windows that would teach the model their relevance. Training on Terra/LUNA corrects this. The secondary classifier now sees depth-withdrawal events at 17% positive density and learns the signature that calm training cannot recover.

CEX market makers withdraw depth under uncertainty regardless of the trigger's cause, because at the order-book timescale they cannot distinguish algorithmic reflexivity from reserve withdrawal from regulatory action. Ask-side depth collapse is therefore a mechanism-agnostic precursor:

Table 19: Mechanism intensity and transfer value.

Training event	Mechanism	Signal	Transfer value
Terra/LUNA	algorithmic	high	strong
Celsius/3AC	exchange credit	medium	useful
FTX	exchange credit	low	near-degenerate
BUSD	regulatory	moderate	useful
Pooled	multi-mechanism	diversified	best

its feature signature transfers across events even as its magnitude differs (5.9× for Terra/LUNA versus 12× for SVB, Table 10). This is the insight that makes cross-event supervision possible without requiring matched training data.

Three regime-detection filters (CUSUM, EWMA, BOCPD [15]) were also evaluated. BOCPD achieves AUROC 0.229 on CEX cross-quote data, in contrast to Cintra and Holloway [9], who detected the Terra depeg 12 hours early from AMM pool reserves. CEX basis is mean-reverting, so changepoint methods cannot substitute for the training-distribution fix that meta-labeling provides.

Figure 8 tests whether training diversity improves oracle capture across four conditions (Terra/LUNA, Celsius/3AC, FTX, and all four pooled), with the threshold calibrated on Terra/LUNA and the test split held fixed on SVB. Oracle capture is stable near 49% for single-event training and rises to 51.8% with four-event pooling (+83.7 bps, 163 trades). The result is therefore about mechanism-diverse transfer rather than generic data augmentation: the training events are selected from the catalogue to span algorithmic/reflexive, exchange-credit/liquidity, and regulatory winddown stress, while fiat-reserve bank shock is held out for testing.

Result 4: mechanism class does not guarantee tradable stress. Mechanism intensity determines whether an event generates executable signal (Table 19). FTX is the informative outlier: its credit shock produced < 5 bps USDC/USDT basis at Binance benchmark venues, so its near-degenerate transfer result reflects weak tradable intensity rather than failure of the exchange-credit class itself. Ask-side depth and spread are the leading predictors in all non-degenerate models.

6.4.2 A Controlled Experiment: Supervision Format, Not Label Density. This is a controlled experiment with one varying factor: how the profitable-window signal is delivered. Meta-labeling receives explicit binary labels (y=1 for every executable window); the conditioned PPO-GRU receives a sparse reward signal over the same windows. Both see identical training data at identical positive-label density (17% of primary-signal minutes are executable). Despite this, the two methods differ by 111.7 bps in net return, isolating supervision format—not density—as the binding constraint.

Without conditioning, the PPO-GRU degenerates to zero entries. The 2.3% positive rate in the full Terra/LUNA stream is too sparse to generate a reward signal that calibrates entry timing. Restricting the agent to primary-signal windows (where the basis threshold fires) resolves the density problem, and the conditioned agent trades 919 windows with AUROC 0.87, yet still earns −29.2 bps.

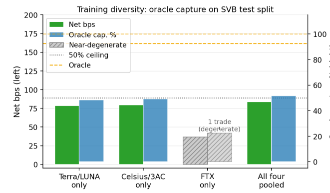

Figure 8: Oracle capture across four training conditions. Dashed: 50% ceiling; hatched: FTX near-degenerate.

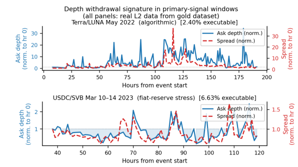

Figure 9: Depth withdrawal signature (real L2). Ask depth (blue) collapses and spread (red) widens identically across Terra/LUNA and SVB stress despite different triggers.

Table 20: RL policy versus supervised baselines (**basis_usdc_1m_gt10bps**, SVB test split).

Model (training)	Net bps	Trades
Oracle (hindsight ceiling)	+161.7	316
Meta-labeling (Terra/LUNA*)	+82.5	397
PPO-GRU conditioned† (Terra/LUNA*)	−29.2	919
PPO-GRU unconditioned (Terra/LUNA*)	degenerate	0
GRU supervised (calm-ctrl)	−239.0	656
MLP-Window supervised (calm-ctrl)	−340.7	1,052

*Cross-mechanism: Terra/LUNA train, SVB eval.
†Conditioned on $|b_{\mathrm{USDC}}|>10$ bps windows, 17% positive rate.

The distinction from meta-labeling is qualitative, not quantitative. Meta-labeling receives an explicit binary label, y=1 for every profitable window, while the RL agent must infer the same distinction through a reward signal that is negative on 83% of entries. Explicit supervision is categorically cleaner than implicit reward.

The overtrading pattern points toward a specific failure mode. The agent fires on 919 of 15,832 test windows, nearly three times the oracle's 316 profitable windows. When a policy fails due to insufficient training, it typically converges toward NoTrade, not toward sustained false-positive entry. The sustained overtrading is therefore more consistent with a timing-transfer mismatch: the agent learned Terra/LUNA's depth-withdrawal sequence and keeps firing on it in the SVB context even when the timing is wrong. Multi-mechanism training on the pooled protocol established here is the direct path to testing whether this mismatch is the binding constraint.

7 LIMITATIONS AND OPEN QUESTIONS

Each limitation below identifies a tractable next step rather than a fundamental ceiling.

Kline-proxy buy leg. The BTC-USDC buy leg uses candlestick-volume depth for the SVB and training event windows because the perpetual contract did not exist on

Binance until January 2024. A 20% depth haircut confirms robustness (§6.2). The proxy dependency resolves directly with 2024 data, where real L2 is already available and partially collected.

Settlement cost assumption. The δ=5 bps penalty is validated against on-chain Etherscan data for 100,000 USDC transfers across the SVB window (Table 21). On nine of ten days the median transfer cost was 1.4–3.2 bps, well below 5 bps. The one exception is March 11 at peak depeg, where gas spiked to 61.5 gwei and the P95 cost reached 8.6 bps—the only day the assumption is tight, and even then the oracle remains profitable at +259.9 bps per trade.

Table 21: On-chain settlement cost validation (USDC ERC-20 transfers, 100K sample, Mar 10–20 2023, ETH ≈ $1,580). δ=5 bps is conservative on all days except the Mar 11 gas spike at peak depeg.

Date	Med gas (gwei)	Med cost (bps)	P95 cost (bps)	Burns
Mar 10	30.8	3.16	5.70	4
Mar 11	61.5	6.31	8.64†	16
Mar 12	21.6	2.21	3.68	22
Mar 13	26.1	2.68	4.47	29
Mar 14–19	13.9–24	1.43–2.46	2.44–4.11	0–2

†Only day where P95 cost exceeds δ=5 bps.
Gas data from 100K USDC transfers via Etherscan API.

Tier-A coverage. Execution-grade claims are currently limited to the SVB stress and recovery windows. Extending to exchange-credit or DeFi events requires route-complete L2 archives not yet available; this is the primary bottleneck for expanding the benchmark's execution-grade scope.

RL market-impact feedback. The conditioned PPO-GRU's −29.2 bps may reflect a timing-transfer mismatch (Terra/LUNA depth-withdrawal sequence firing at the wrong moments in SVB) rather than a fundamental RL ceiling. Pooled multi-mechanism training on the protocol established here is the direct experiment to test whether this mismatch explains the gap. Additionally, training on historical replay without market-impact feedback miscalibrates sizing in thin-book regimes, because the agent sees depth as it was rather than as it would be after its own orders clear liquidity. Integrating the cross-mechanism initialisation protocol into a reactive simulator resolves both issues simultaneously and is the natural next step toward closing the 111.7 bps gap between meta-labeling and the RL agent.

8 CONCLUSION

The 12× optical-to-executable gap shows that stablecoin arbitrage benchmarks built on price labels measure the wrong quantity. Executable opportunities are scarce, shrink with trade size, and are invisible to any model trained on calm-market data. The only empirically validated path to positive returns is cross-mechanism training on a structurally different stress event, and the RL diagnostic explains precisely why: label density is necessary for the model to trade at all, but explicit binary supervision over the profitable subset is what produces the directional accuracy that reward optimisation cannot replicate at low positive rates. The released benchmark includes ingestion, normalization, book reconstruction, features, labels, models, and evaluation under `src/stressbench`; SQL schemas; notebooks; tests; the machine-readable event windows in `configs/event_windows_historical.yaml`; the source verification registry in `src/stressbench/history/source_verification.py`; executable labels; oracle implementation; model scripts; and paper tables. StressBench supports five future directions: (1) multi-mechanism RL training on the pooled-event protocol established here, the direct path to closing the 111.7 bps gap between meta-labeling and the conditioned RL agent; (2) execution-aware detection under cost uncertainty; (3) cross-mechanism transfer to new event types as Tier-A coverage expands; (4) RL policy learning with the cross-mechanism initialisation protocol established here; and (5) reactive simulation of thin-book markets, where the oracle ceiling provides a principled performance target. The benchmark harness additionally ships four interpretable single-feature baselines that exercise the same evaluation pipeline and serve as reference points for future model comparisons. The object we want future work to remember is therefore not an SVB case study, but an 18-event historical stress universe narrowed to two execution-grade windows and a cross-mechanism transfer ceiling.

A FULL HISTORICAL EVENT CATALOGUE

Table 22 reports event-level metadata used by the main paper. All Tier-B magnitudes are price/liquidity-grade or configured estimates, and Tier-C rows are taxonomy-only.

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Table 22: Appendix A: full historical event catalogue.

Event ID	Date	Stablecoin	Mechanism	Tier	Cov.	Verif.	Max depeg	Benchmark use
dai_black_thursday_2020	Mar 2020	DAI	collateral/liquidation	B	0.50	verified	+150 bps est.	above-peg stress
fei_launch_2021	Apr 2021	FEI	algorithmic/reflexive	C	0.25	needs source	-2000 bps est.	taxonomy
iron_titan_2021	Jun 2021	IRON	algorithmic/reflexive	C	0.25	needs source	-10000 bps est.	taxonomy
mim_wonderland_2022	Jan 2022	MIM	algorithmic/reflexive	C	0.25	needs source	-300 bps est.	taxonomy/context
terra_ust_2022	May 2022	UST	algorithmic/reflexive	B	0.50	verified	-10000 bps est.	validation, transfer
usdd_tron_2022	Jun 2022	USDD	algorithmic/reflexive	B	0.50	needs source	-200 bps est.	taxonomy
celsius_3ac_2022	Jun 2022	USDT/USDC	exchange credit/liquidity	B	0.50	needs source	-100 bps est.	transfer
husd_depeg_2022	Aug 2022	HUSD	exchange credit/liquidity	B	0.50	needs source	-800 bps est.	taxonomy
acala_ausd_2022	Aug 2022	aUSD	RWA/niche	C	0.25	needs source	-9900 bps est.	taxonomy
binance_stablecoin_conversion_2022	Sep 2022	BUSD/USDC	regulatory wind-down	C	0.25	needs source	0 bps	context
ftx_collapse_2022	Nov 2022	USDT/USDC	exchange credit/liquidity	B	0.50	verified	-50 bps est.	transfer contrast
busd_regulatory_2023	Feb 2023	BUSD	regulatory wind-down	B	0.50	verified	-30 bps est.	transfer
usdc_svb_2023	Mar 2023	USDC	fiat-reserve bank shock	A	1.00	verified	-1300 bps	execution benchmark
usdc_svb_recovery_2023	Mar 2023	USDC	fiat-reserve bank shock	A	0.75	verified	-10 bps	recovery benchmark
curve_3pool_ust_2022	May 2022	UST/3CRV	DeFi pool imbalance	B	0.50	needs source	-500 bps est.	on-chain stress
usdc_dai_secondary_svb_2023	Mar 2023	DAI/FRAX	DeFi pool imbalance	B	0.75	needs source	-200 bps est.	DeFi contagion
usdt_curve_2023	Jun 2023	USDT	DeFi pool imbalance	B	0.50	verified	-80 bps est.	on-chain stress
usdr_2023	Oct 2023	USDR	RWA/niche	B	0.50	needs source	-5000 bps est.	long-tail stress

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